

ECON485 Fall 2025 – Homework Assignment 1

Course Registration System – Database Design Homework
Appendix 1 is required for this homework.

See Appendix 1: University Course Registration Rules and Regulations for the full description of the system.

Part 1 – Extracting Concepts (Business Objects)

Read Appendix 1 carefully. Your first task is to extract the important concepts (also called business objects or entities) that appear in the text.

Task 1:

Create a list of all concepts that you think are important for a “course registration system.” Examples of concepts include: Course, Student, Instructor, Section, etc. Your list should include both basic concepts and more advanced concepts, such as prerequisites, grade replacement, withdrawals, and so on. You do not need to define their properties in this part. A simple bullet list is enough.

Part 2 – Selecting the Basic Concepts (Minimum System Without Rules)

For this part, imagine a very simple course registration system with:

- No restrictions on what students can register for
- No prerequisites or co-requisites
- No limits on credits
- No rules about time conflicts
- No add/drop or withdrawal history
- No repeated course tracking

We only need the basic features that allow:

- Creating courses
- Creating students
- Offering sections of courses
- Registering students to sections

Task 2:

From your list in Part 1, choose only the concepts needed for this very simple system. Make a list of the selected concepts. These concepts will be used in Part 3.

Part 3a – 2NF Design and ER Diagram (Basic System Only)

In this part, you design a 2NF (Second Normal Form) database for the basic system described in Part 2.

Task 3a:

Create tables for each concept you selected in Part 2. Show the fields (columns) that you think each table needs.

Make sure your design follows 2NF rules:

- No repeating groups
- No partial dependencies on a composite key

Draw an ER diagram showing:

- Tables
- Primary Keys
- Foreign Keys
- Relationships (1-N, N-M if needed)

You may draw the ER diagram by hand or using a simple tool. The drawing does not need to be very artistic, only clear.

Part 3b – Explain the Relations and Keys

Task 3b:

Write a number of short explanation (max. 5–10 sentences each) describing:

- Why you chose your primary keys
- Why you chose your foreign keys
- How the tables are connected
- Any N-M relationships and how you solved them
- Why the design satisfies 2NF

Keep the explanations simple and clear.

Part 4a – Constraints That Force a Student to Register for a Course (Prerequisites, co-requisites, required chains)

In Appendix 1, some rules require a student to take a course before or together with another course. These rules influence the database design.

Examples:

- Prerequisites must be passed before taking a course.
- Co-requisites must be taken in the same semester.

- Some courses must be repeated if failed.

Task 4a:

Write a short discussion (8–12 sentences) explaining:

- Why these constraints are important in the real system
- What new tables or fields would be needed to support them
- (for example: a “Prerequisite” table, or a “MinimumGrade” field)
- What information must be stored to check these constraints
- Why these tables/fields are not required in the simple system from Part 2

You do not need to draw ER diagrams or design full tables. Only verbal discussion is required.

Part 4b – Constraints That Limit What Students Can Register For (Time conflicts, maximum credit load, mutually exclusive courses, etc.)

Appendix 1 also includes several rules that block students from registering for certain courses.

Examples:

- Time conflicts between course sections
- Upper limits on ECTS credits
- Mutually exclusive courses
- Withdrawal or failure rules that block registration

Task 4b:

Write a short discussion (8–12 sentences) explaining:

- Why these limiting rules matter in a real system
- What extra tables or fields are needed to enforce them
- (for example: a “Schedule” table or a “MutualExclusion” table)
- What information the system needs to check before allowing registration
- Why these rules were removed from the basic system in Part 2

Again, no diagrams or full table designs are required. Only verbal discussion.

Part 4c – The Requirement to Keep History

Appendix 1 mentions that the real system keeps a history of all actions, such as:

- Add/drop attempts
- Withdrawal attempts
- Overrides
- Changes in sections
- Forced withdrawals
- Time conflict approvals

A basic database does not have this feature.

Task 4c:

Write a short discussion (8–12 sentences) explaining:

- Why a real registration system needs to keep a history of actions
- What extra tables or fields would be needed to store history (for example: an “ActionLog” table with timestamp, type of action, etc.)
- Why this requirement makes the system more complex
- Why we did not include this history in the simplified design of Part 3

No diagram is needed here.